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Department of Information Technology
Academic Year 2024 - 2025 (Even Semester)

Semester, Degree & Branch: VI Semester, B.Tech. IT
Course Code & Title: CCS335 & Cloud Computing
Name of the Faculty member (s): Mrs. A. Alagulakshmi, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit II / Virtualization of memory and I/O devices
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO2
- **Activity Chosen:** Learning by Teaching
- **Justification:**

  Understanding the virtualization of memory and I/O devices is critical for students in Unit II of the Cloud Computing course. This activity allows students to gain deeper insights into these concepts through a peer-led teaching approach.

- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**

  After the lecture session, students were asked to write about the foundations of the design thinking process. Following this, in the final 20 minutes, Mr. P. Siva Sekar conducted a peer-teaching session. **First 10 minutes:** Introduction to virtualization. **Next 10 minutes:** Discussion on memory and I/O device virtualization in distributed computing systems. **Figure 1** illustrates the implementation of this learning-by-teaching activity.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO5	PO9	PO11	PO12	PSO3
CO2	3	2	2	3	2	2	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO5**	**PO9**	**PO11**	**PO12**	**PSO3**
	3	2	2	3	2	2	3
Justification for correlation	Students can apply IT principles to solve issues related to virtualization.	Students assess different virtualization techniques and understand limitations.	Students identify and use modern tools and resources for creating virtual machines.	Students develop communication and problem-solving skills by presenting on virtualization.	Students evaluate the impact of virtualization in engineering contexts.	Students realize the importance of lifelong learning through the evolving cloud technologies.	Students develop critical thinking to foresee challenges in virtualization and propose effective solutions.

- **Images / Screenshot of the practice:**

Figure 1. Learning by Teaching by Sivasekar P

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
    - Students demonstrated better comprehension of memory and I/O device virtualization.
    - Recognized the importance and application of virtualization in cloud computing.
    - Gained hands-on experience with presenting technical concepts.
- ***Benefit of the practice:***
    - Encouraged active student participation.
    - Improved conceptual clarity through peer explanation.
    - Promoted collaboration and knowledge sharing.
- ***Challenges faced in implementation:***
    - The presenting student initially lacked confidence and struggled with clarity.
    - The volume of content made it challenging to manage time effectively within the session.

References:

- https://corp.kaltura.com/blog/innovative-teaching
- https://www.youtube.com/watch?v=wz9JLHODQ74
- https://www.youtube.com/watch?v=OI_3bQ-EWSI

Signature of Faculty Member **HOD**